WORKED SOLUTION

STEP II 2025, Question 6

Pure • Circles Touching a Parabola

STEP II 2025, Question 6

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2025 STEP papers (a zip file):

www.ocr.org.uk/Images/750486-2025-past-papers.zip

The key idea

Every circle here is centred on the y-axis, so everything is symmetric in x and "touching at two points" means touching at a mirror pair $(\pm x_0, y_0)$ with $x_0 \neq 0$. Tangency is easiest to capture by saying the normals agree: the radius to the point of contact must be along the normal to the other curve. Once parts (i) and (ii) each give a formula for r^2 in terms of the centre height a, part (iii) is about a quadratic in a whose two roots are a_1 and a_2.

Solution

Part (i)

Suppose the circle touches the parabola at $P = (x_0, y_0)$, with $x_0 \neq 0$. (Two distinct points of contact cannot both lie on the axis, since the parabola meets $x = 0$ only at the origin; and by symmetry, if (x_0, y_0) is a point of contact then so is $(-x_0, y_0)$.)

The normal to the circle at P is along the radius, $(x_0,\ y_0 - a)$. Writing the parabola as $2ky - x^2 = 0$, its normal is along the gradient $(-2x_0,\ 2k)$. Tangency means these are parallel:

$$x_0 \cdot 2k - (y_0 - a)(-2x_0) = 0 \quad \Longrightarrow \quad 2x_0\,(k + y_0 - a) = 0.$$

Since $x_0 \neq 0$, we get $y_0 = a - k$. Then P is on the parabola, so $x_0^2 = 2ky_0 = 2k(a-k)$, and P is on the circle, so

$$r^2 = x_0^2 + (y_0 - a)^2 = 2k(a-k) + k^2 = k(2a-k),$$

as required.

Conversely, let $a > k > 0$ and $r^2 = k(2a-k)$. Put $y_0 = a - k > 0$ and $x_0 = \sqrt{2k(a-k)} > 0$. Both $(\pm x_0, y_0)$ lie on the parabola by construction, and on the circle because $x_0^2 + (y_0 - a)^2 = 2k(a-k) + k^2 = r^2$. At each of these points the normals $(\pm x_0, -k)$ and $(\mp 2x_0, 2k)$ are parallel, so the curves share a tangent there. Hence the circle touches the parabola at two distinct points, since $x_0 \neq 0$.

(As a cross-check, substituting $x^2 = 2ky$ into the circle gives $y^2 - 2(a-k)y + (a-k)^2 = 0$, that is $(y-(a-k))^2 = 0$: a repeated root, which is the algebraic signature of tangency.)

Part (ii)

The line $y = c + x$, or $x - y + c = 0$, is tangent exactly when its distance from the centre $(0, a)$ equals r:

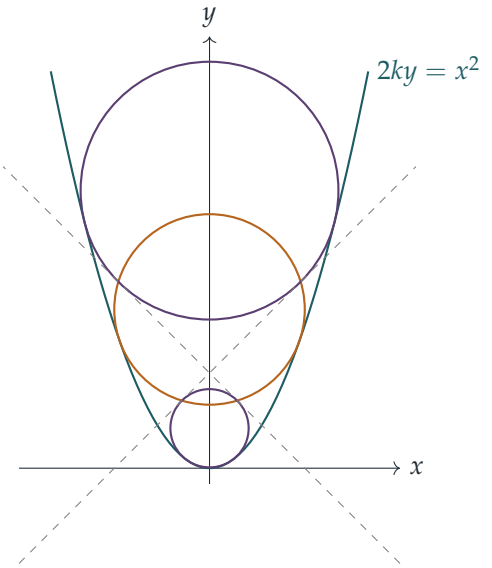
$$r = \frac{|0 - a + c|}{\sqrt{2}} \quad \Longrightarrow \quad r^2 = \frac{(a-c)^2}{2}.$$

The line $y = c - x$ is the mirror image in the y-axis, so it gives the same condition.

The point of contact is the foot of the perpendicular from $(0, a)$. The radius is perpendicular to the line, so it has direction $(1, -1)$; moving from $(0, a)$ by $t(1, -1)$ and landing on $y = c + x$ gives $a - t = c + t$, so $t = \frac{a-c}{2}$. The point of contact with $y = c + x$ is

$$\left(\tfrac{a-c}{2},\ \tfrac{a+c}{2}\right),$$

and by symmetry the point of contact with $y = c - x$ is $\left(-\tfrac{a-c}{2},\ \tfrac{a+c}{2}\right)$.



The case $k = 1$, $c = 3$: the parabola, the two circles C_1, C_2 (plum) touching it and the lines $y = 3 \pm x$ (dashed), and the circle of part (iii)(b) (amber).

Part (iii)(a)

Each circle touches the parabola at two points, so by part (i), $r_i^2 = k(2a_i - k)$; and each is tangent to the lines $y = c \pm x$, so by part (ii), $r_i^2 = \frac{1}{2}(a_i - c)^2$. Equating,

$$(a_i - c)^2 = 2k(2a_i - k) \quad \Longrightarrow \quad a_i^2 - 2(c + 2k)\, a_i + c^2 + 2k^2 = 0.$$

So a_1 and a_2 both satisfy the quadratic $a^2 - 2(c+2k)a + c^2 + 2k^2 = 0$. As $a_1 \neq a_2$, they are its two roots, and so

$$a_1 + a_2 = 2c + 4k, \qquad a_1 a_2 = c^2 + 2k^2.$$

Then

$$\begin{aligned} a_1^2 + a_2^2 &= (a_1 + a_2)^2 - 2a_1 a_2 \\ &= 4c^2 + 16kc + 16k^2 - 2c^2 - 4k^2 \\ &= 2c^2 + 16kc + 12k^2, \end{aligned}$$

which completes (a).

Part (iii)(b)

By part (ii), the points of tangency on C_i are $(\pm\frac{a_i - c}{2},\ \frac{a_i + c}{2})$. It is tidier to write $u_i = a_i - c$, so the points are $(\pm\frac{u_i}{2},\ c + \frac{u_i}{2})$.

> **Ask yourself** We need one circle through four points, but only have two unknowns d and p. How can we pin them down without solving for a_1 and a_2 explicitly?

The four points are two mirror pairs, and any circle centred on the y-axis is symmetric, so it is enough to make the circle pass through $(\frac{u}{2}, c + \frac{u}{2})$ for $u = u_1$ and $u = u_2$. This condition is

$$\frac{u^2}{4} + \left(c - d + \frac{u}{2}\right)^2 = p^2 \quad \Longleftrightarrow \quad u^2 + 2(c-d)\,u + 2(c-d)^2 - 2p^2 = 0. \qquad (*)$$

On the other hand, substituting $a = u + c$ into the quadratic from (a) gives

$$u^2 - 4k\,u + 2k^2 - 4kc = 0, \qquad (**)$$

whose roots are exactly u_1 and u_2, which are distinct because $a_1 \neq a_2$. The circle passes through all four points precisely when u_1 and u_2 are both roots of $(*)$. Two monic quadratics with the same two distinct roots are identical, so we compare coefficients with $(**)$:

$$2(c-d) = -4k \quad \Longrightarrow \quad d = c + 2k,$$

and then $2(c-d)^2 - 2p^2 = 8k^2 - 2p^2 = 2k^2 - 4kc$, so

$$p^2 = 3k^2 + 2kc = k(3k + 2c).$$

Part (iii)(c)

We use the converse in part (i). With $d = c + 2k$,

$$k(2d - k) = k(2c + 3k) = p^2,$$

so the circle has exactly the form $r^2 = k(2a - k)$, with d in place of a. It remains to check that $d > k$.

Ask yourself Where does an inequality like $d > k$ come from, when nothing in (a) or (b) mentioned one?

It comes from the fact that C_1 and C_2 really do touch the parabola at two points. In part (i), touching at two points forced the points of contact to have $y_0 = a - k$ with $x_0^2 = 2k(a-k) > 0$ (as $x_0 \neq 0$), so $a_1 > k$ and $a_2 > k$. Adding these and using (a),

$$2c + 4k = a_1 + a_2 > 2k \quad \Longrightarrow \quad c > -k \quad \Longrightarrow \quad d = c + 2k > k.$$

So $d > k > 0$ and $p^2 = k(2d-k)$, and by the second half of part (i) the circle $x^2 + (y-d)^2 = p^2$ touches the parabola $2ky = x^2$ at two points, namely $(\pm\sqrt{2k(c+k)},\ c+k)$.

Checks

Take $k = 1$, $c = 3$ (the picture above). The quadratic in (a) is $a^2 - 10a + 11 = 0$, so $a = 5 \pm \sqrt{14} \approx 8.742$ and 1.258. Their sum is $10 = 2c + 4k$ and their sum of squares is $78 = 2 \cdot 9 + 16 \cdot 3 + 12$. Part (b) gives $d = 5$ and $p^2 = 9$. The circle $x^2 + (y-5)^2 = 9$ meets $2y = x^2$ where $2y + (y-5)^2 = 9$, that is $(y-4)^2 = 0$: a double root at $y = 4$, so it touches at $(\pm 2\sqrt{2}, 4)$, as (c) predicts. Each of the four points of tangency, for instance $(\frac{\sqrt{14}+2}{2}, \frac{\sqrt{14}+8}{2})$, lies at distance 3 from $(0,5)$. All the results were also confirmed by computer for hundreds of random admissible k and c.

Common slips

- In (i), forgetting that the touching points have $x_0 \neq 0$. That is what allows us to divide by x_0, and it is also what gives $a > k$.
- In (iii)(a), using only $r^2 = k(2a-k)$ or only $r^2 = \frac{1}{2}(a-c)^2$. The quadratic for a comes from needing both at once.
- In (iii)(c), checking $p^2 = k(2d-k)$ but not $d > k$. Without it the "circle" might touch only at the origin, or not at all.

Beyond STEP

By part (i), the circles centred on the y-axis that touch the parabola twice are $x^2 + y^2 - k^2 - 2a(y-k) = 0$: an equation *linear* in the parameter a. Such a family is called a pencil (or coaxal system) of circles, and C_1, C_2 and the circle in (b) all belong to it. Matching normals is the first step towards the idea of *contact* between curves; the circle that touches a curve to second order is its circle of curvature, met in a first course on the differential geometry of curves. Siklos, *Advanced Problems in Mathematics*, has many more STEP-style coordinate geometry problems in this spirit.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.